

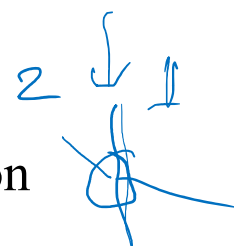
Heat Transfer Fundamentals

Rate (Heat IN) – (Heat OUT) ± Rate of Generation = Rate of Accumulation

Fourier's Law $q_x = -k A \frac{dT}{dx}$ Newton's Law of Cooling $q = h A (T_s - T_\infty)$

Cartesian coordinates (x, y, z)

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi_v + \dot{Q}$$

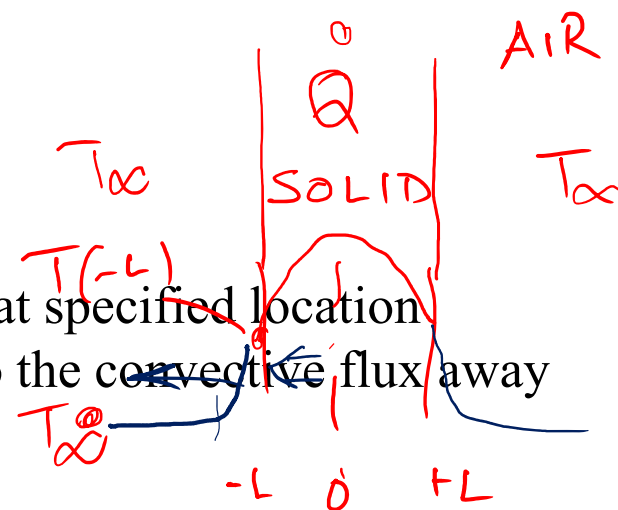


For a system experiencing only conduction and no viscous heat dissipation

$$\rho \hat{c}_p \left(\frac{\partial T}{\partial t} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \dot{Q}$$

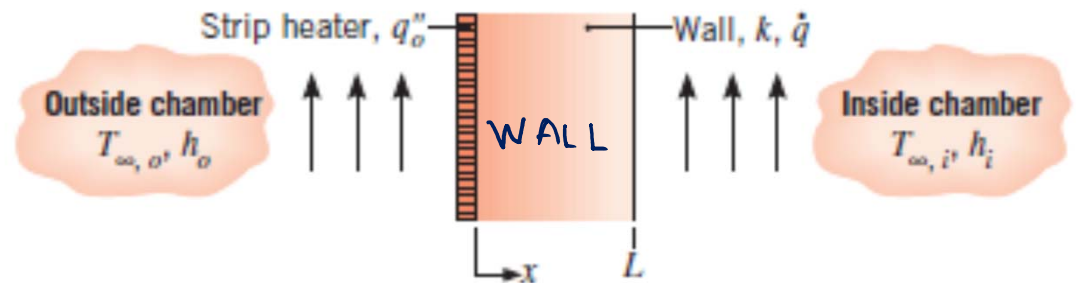
Possible boundary conditions:

1. Temperature specified at a point $T(x)$ is known
2. Equality of temperature at the interface $T_{s1} = T_{s2}$
3. Perfect insulation and/or adiabatic surface $dT/dx = 0$ at specified location
4. Conductive flux to the solid-fluid interface is equal to the convective flux away from the interface, $-k \frac{dT}{dx} \big|_{x=L} = h(T_s - T_\infty)$

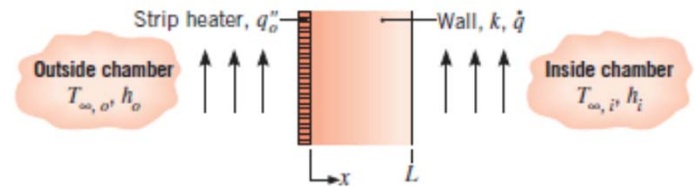


Wall Heat Generation with Strip Heater Experiencing Convection at the Boundaries

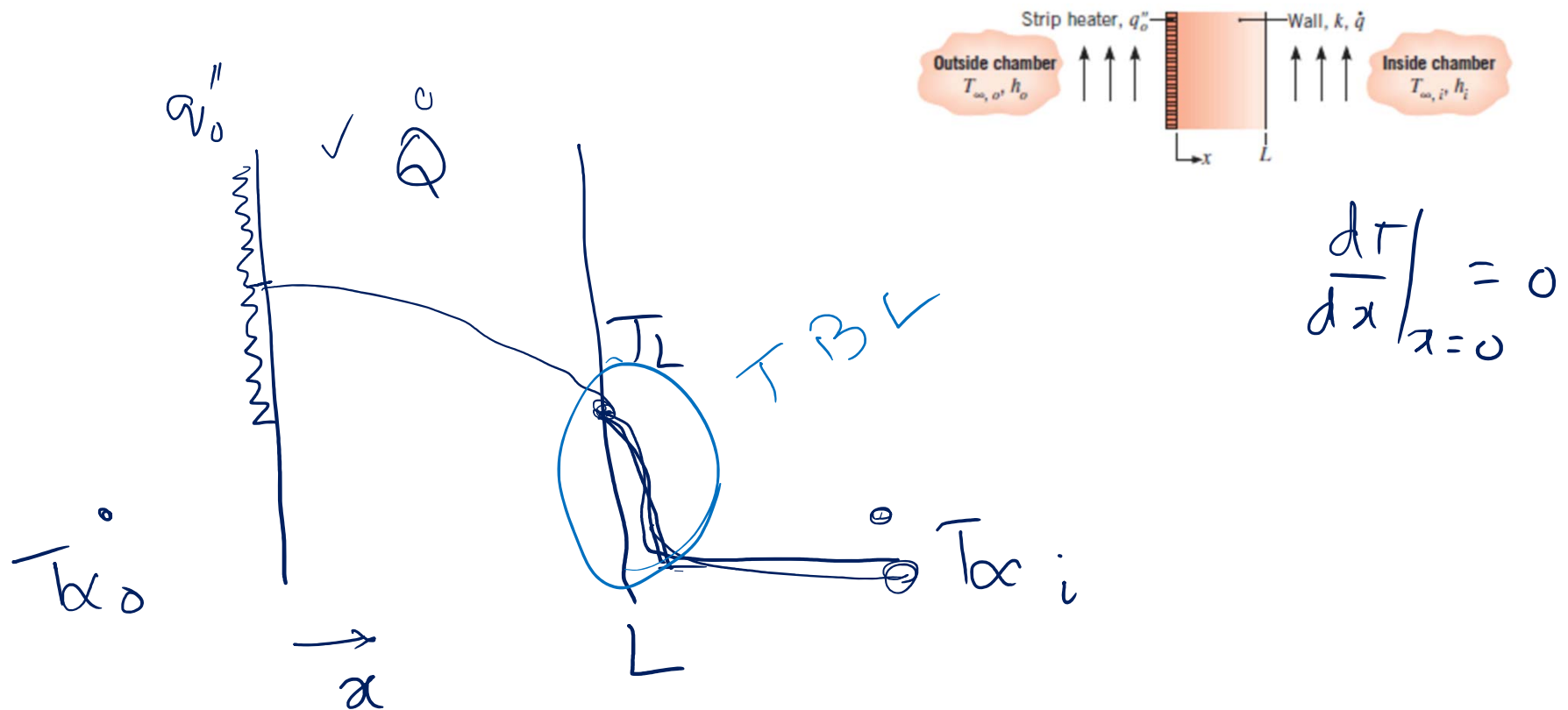
The air *inside* a chamber at $T_{\infty,i} = 50^\circ\text{C}$ is heated convectively with $h_i = 20 \text{ W/m}^2 \cdot \text{K}$ by a 200-mm-thick wall having a thermal conductivity of $4 \text{ W/m}\cdot\text{K}$ and a uniform heat generation of 1000 W/m^3 . To prevent any heat generated within the wall from being lost to the *outside* of the chamber at $T_{\infty,o} = 25^\circ\text{C}$ with $h_o = 5 \text{ W/m}^2 \cdot \text{K}$, a very thin electrical strip heater is placed on the outer wall to provide a uniform heat flux, q''_o



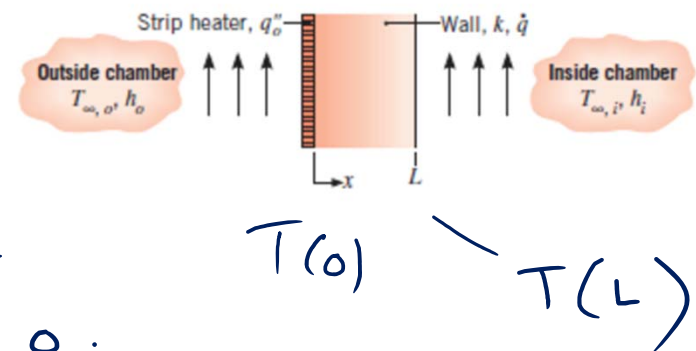
- (a) Sketch the T-distribution in the wall on $T - x$ coordinates for the condition where no heat generated within the wall is lost to the *outside* of the chamber.
- (b) What are the temperatures, $T(0)$ and $T(L)$, for the conditions of part (a)?
- (c) Determine the value of q''_o that must be supplied by the strip heater so that all heat generated within the wall is transferred to the *inside* of the chamber.
- (d) If the heat generation in the wall were switched off while the heat flux to the strip heater remained constant, what would be the steady-state temperature, $T(0)$, of the outer wall surface?



Sketch the T-distribution in the wall on $T - x$ coordinates for the condition where no heat generated within the wall is lost to the *outside* of the chamber.



What are the temperatures, $T(0)$ and $T(L)$, for the conditions of part (a) i.e. for the condition where no heat generated within the wall is lost to the *outside* of the chamber?



$$\frac{d^2 T}{dx^2} + \frac{\dot{q}_o}{k} = 0.$$

$$\Rightarrow T = -\frac{\dot{q}_o x^2}{2k} + C_1 x + C_2$$

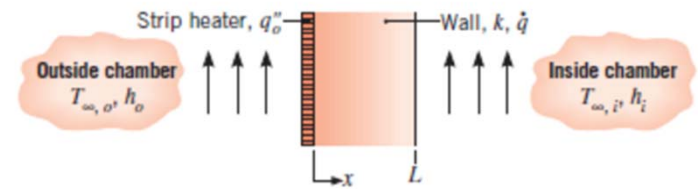
$$dT/dx = 0 \text{ at } x=0 \rightarrow C_1 = 0.$$

$$-k dT/dx|_{x=L} = h_i (T_L - T_{i,i})$$

$$\dot{q}_o L = h_i (T_L - T_{i,i}) \Rightarrow T_L = \frac{\dot{q}_o L}{h_i} + T_{i,i}$$

$$T - T_L = \frac{\dot{q}_o L^2}{2k} \left[1 - \left(\frac{x}{L} \right)^2 \right] \Rightarrow T|_{x=0} = T_0 = 65^\circ \text{C}$$

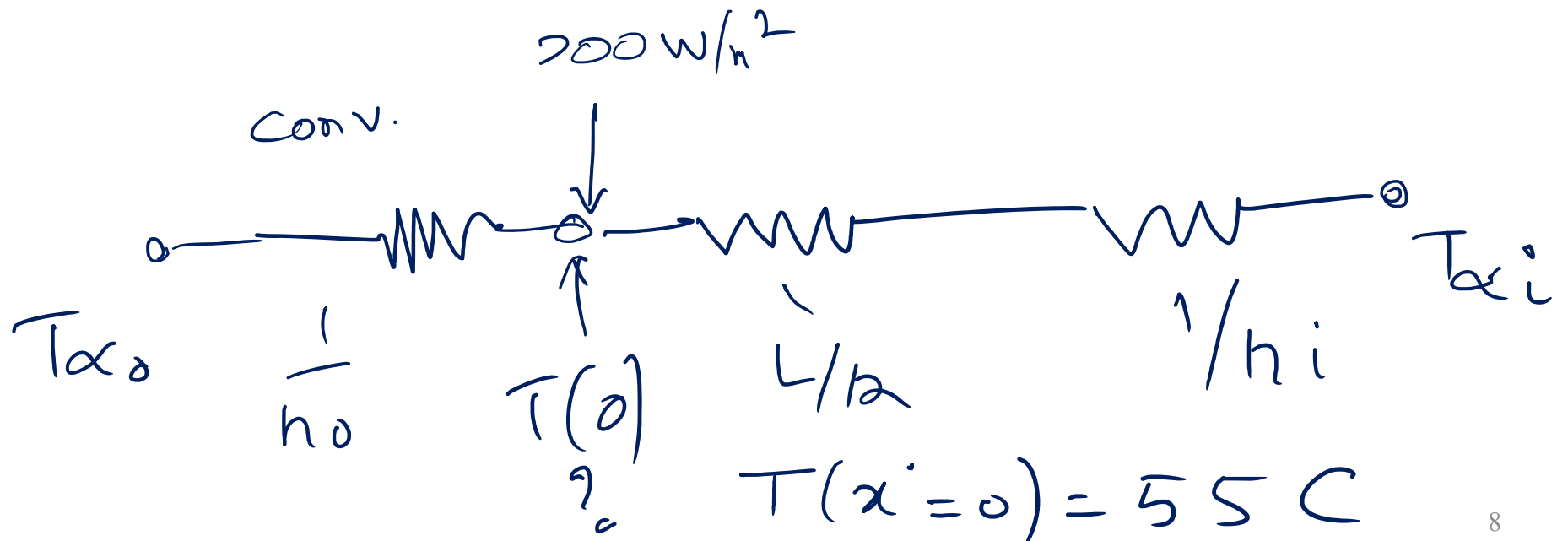
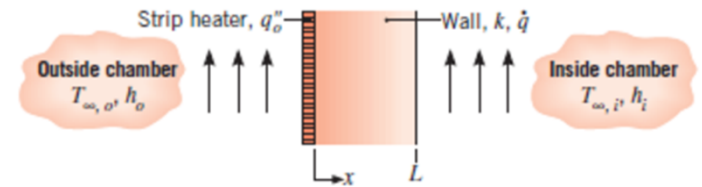
Determine the value of q''_0 that must be supplied by the strip heater so that all heat generated within the wall is transferred to the *inside* of the chamber.



$$q''_0 = h_o (T|_{x=0} - T_o)$$

$$= 5 (65 - 25) = 200 \text{ W/m}^2$$

If the heat generation in the wall were switched off while the heat flux to the strip heater remained constant, what would be the steady-state temperature, $T(0)$, of the outer wall surface?



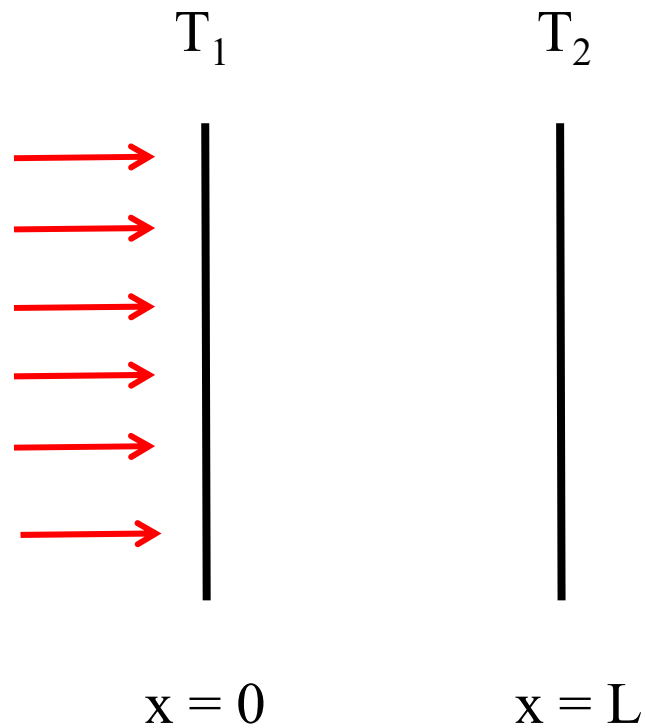
Problem 2 – Heat Generation in a Plane Wall

Consider a plane wall of thickness L , which is to act as shielding for a nuclear reactor. The inner surface receives gamma radiation that is partially absorbed within the shielding and has the effect of an internally distributed heat source. In particular heat is generated per unit volume within the shielding according to the relation

$$q(x) = q_o'' \alpha e^{-\alpha x}$$

Where q_o'' is the incident radiation flux and α is the absorption coefficient of the shielding.

- (a) If the inner ($x = 0$) and outer ($x = L$) surfaces of the shielding are maintained at constant temperatures of T_1 and T_2 respectively, find the temperature distribution in the shielding?
- (b) Find an expression for the x location in the shield at which the temperature is a maximum.



Distributed heat source

$$q(x) = q_o'' \alpha e^{-\alpha x}$$

$q(x)$ = heat generation per
unit volume

1 – Dimensional SS conduction

Governing equation

$$0 = k \left[\frac{d^2 T}{dx^2} \right] + \dot{q}$$

Gov. Eqn, $0 = k \left[\frac{d^2 T}{dx^2} \right] + \dot{q}$

$$k \frac{d^2 T}{dx^2} = -q_o'' \alpha e^{-\alpha x} \longrightarrow \frac{dT}{dx} = \frac{A}{-\alpha} e^{-\alpha x} + C_1 \quad A \equiv -\frac{q_o'' \alpha}{k}$$

Therefore

$$T = -\frac{q_o''}{\alpha k} e^{-\alpha x} + C_1 x + C_2$$

Using BC 1 at $x = 0$, $T = T_1 \longrightarrow C_2 = T_1 + \frac{q_o''}{\alpha k}$

$$T - T_1 = \frac{q''_o}{\alpha k} (1 - e^{-\alpha x}) + C_1 x$$

Using BC 2 at $x = L$ $T = T_2$

$$C_1 = \frac{1}{L} \left[(T_2 - T_1) - \frac{q''_o}{\alpha k} (1 - e^{-\alpha L}) \right]$$

Therefore

$$T = T_1 - (T_1 - T_2) \frac{x}{L} + \frac{q''_o}{\alpha k} \left(1 - \frac{x}{L} \right) + \frac{q''_o}{\alpha k} \left(\frac{x}{L} e^{-\alpha L} - e^{-\alpha x} \right)$$

Location at which T is maximum

$$\frac{dT}{dx} = (T_2 - T_1) \frac{1}{L} + \frac{q''_o}{k} e^{-\alpha x} - \frac{q''_o}{\alpha k L} (1 - e^{-\alpha L})$$

$\frac{dT}{dx} = 0$ gives

$$e^{-\alpha x} = (T_2 - T_1) \frac{k}{L q''_o} + \frac{q''_o}{\alpha k L} (1 - e^{-\alpha L})$$

Problem 3 - Heater Sphere in Quiescent Fluid

A heated sphere of radius R is suspended in a large, motionless body of fluid. It is desired to study the heat conduction in the fluid surrounding the sphere in the absence of convection. The surface temperature of the sphere is maintained at T_R and at a point far from the sphere, the temperature is T_∞ . Obtain the functional relationship of the temperature of the surrounding fluid as a function of r , the distance from the centre of the sphere. The thermal conductivity of the fluid, k , is constant. Using the above relation and the Newton's law of cooling evaluate an expression of Nusselt number for this case.

Spherical coordinates (r, θ, ϕ)

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = k \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + \mu \phi_v + \dot{Q}$$

$\tau(r)$

Using the above relation and the Newton's law of cooling evaluate an expression of Nusselt number for this case.

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

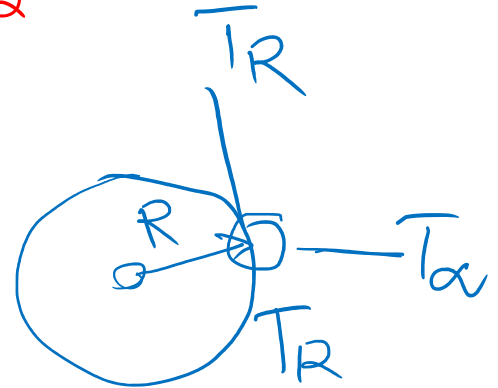
$$\frac{1}{r^2} \frac{d}{dr} \left(k r^2 \frac{dT}{dr} \right) = 0 \quad \text{Gov. EQN}$$

$$r^2 \frac{dT}{dr} = C_1$$

$$T = -\frac{C_1}{r} + C_2$$

$$r \rightarrow \infty \quad T \rightarrow T_\infty$$

$$r = R \quad T = T_R$$



$$Nu \equiv \frac{hD}{k}$$

$$\frac{T - T_\infty}{T_R - T_\infty} = \frac{R}{r} \quad \checkmark$$

AIR

T(r)

$$\frac{T - T_{\infty}}{T_R - T_{\infty}} = \frac{R}{r}$$

AT THE INTERFACE

$$Nu = 2$$

$$q_{\text{COND.}} = q_{\text{CONV.}}^{\checkmark}$$

$$-k \frac{dT}{dr} \Big|_{r=R} = h(T_R - T_{\infty})$$

$$\frac{hR}{k} = 1$$

$$Nu = \frac{hD}{k} = 2$$

Nu

$$Nu = 2.0 + f(Re)$$

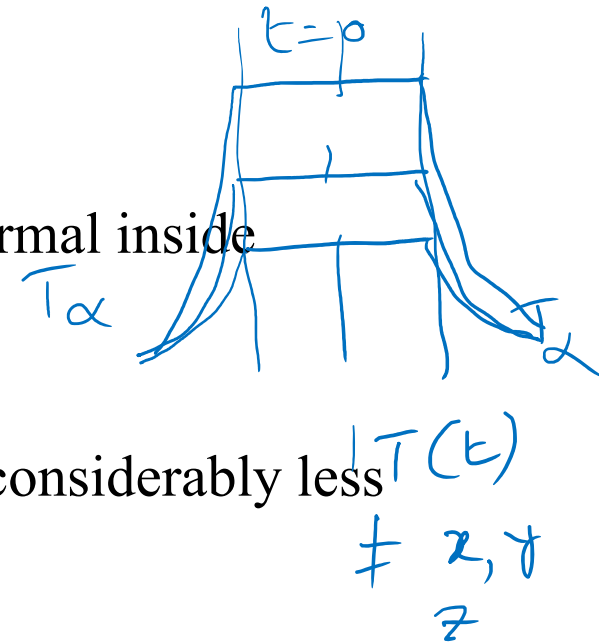
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Transient Conduction

Lumped Capacitance Method

Major assumption: Temperature is space-wise isothermal inside the solid at any instant of time

Valid if the conduction resistance inside the solid is considerably less than the convective resistance at the periphery



$$\text{Biot Number } (= hL/k) \ll 1$$

$$\dot{In} - \dot{Out} + \dot{Generation} = \dot{Stored}$$

For a hot solid being cooled by a convective process, e.g. by putting it in a cold liquid and no heat generation

$$- \dot{E}_{OUT} = \dot{E}_{STORED}$$

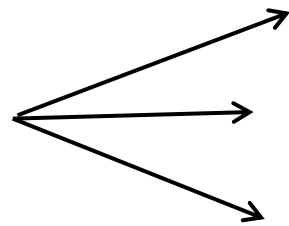
$$- h A_s (T - T_{\infty}) = \rho V C \frac{dT}{dt}$$

$$\frac{\rho V C}{h A_s} \frac{d\theta}{dt} = - \theta$$

$$\frac{\rho V C}{h A_s} \ln \frac{\theta_i}{\theta} = t$$

$$\frac{\theta}{\theta_i} = \frac{T - T_{\infty}}{T_i - T_{\infty}} = \exp \left[- \left(\frac{h A_s}{\rho V C} \right) t \right]$$

Alternative to
LC Method



Analytical

Numerical

Graphical (Heisler Charts)

Problem on Transient Conduction

A long wire of diameter $D = 1 \text{ mm}$ is submerged in an oil bath of temperature $T_{\infty} = 25^{\circ}\text{C}$. The wire has an electrical resistance per unit length of $R_e' = 0.01 \text{ ohm/m}$. A current of $I = 100 \text{ A}$ flows through the wire and the convection coefficient is $h = 500 \text{ W/m}^2\cdot\text{K}$.

What is the steady state temperature of the wire?

From the time the current is applied, how long does it take for the wire to reach a temperature within 1°C of the steady state value?

The properties of the wire are $\rho = 8000 \text{ kg/m}^3$, $c = 500 \text{ J/kg}\cdot\text{K}$ and $k = 20 \text{ W/m}\cdot\text{K}$

The first step is to calculate Biot number.

$$Bi \equiv \frac{h(x_0)}{k} = \frac{500 \frac{W}{m^2 K} \cdot 5 \times 10^{-4} m}{20 W/m K} = 0.012 < 0.1$$

As $Bi < 0.1$, LC method is applicable

($T \neq f(x)$, $T = f(t)$ only)

At steady state

In - Out + Generation = Stored

$$- \dot{O}UT + \dot{G}EN = 0$$

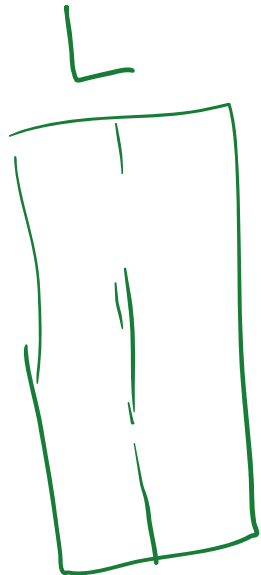
$$\pi (h D L) (T - T_\infty) = I^2 R'_e L$$

$R'_e = \text{Resistance/length}$

$$T = T_\infty + \frac{I^2 R'_e}{\pi D h}$$

$$= 25 + \frac{(100)^2 \times 0.01}{\pi \times 0.001 \times 500} = 88.7^\circ C$$

$\therefore$ Steady state temperature = $88.7^\circ C$



$h/2$

$h/2$

For the transient case,

$$\dot{Q}_{IN} - \dot{Q}_{OUT} + \dot{Q}_{GEN} = \dot{Q}_{STORED}$$

$$0 - h\pi D(T - T_{\infty}) + I^2 R_e' = \rho C \frac{\pi D^2}{4} \frac{dT}{dt}$$

$$\frac{dT}{dt} = \frac{I^2 R_e'}{\rho C \pi D^2 / 4} - \frac{4h}{\rho C D} (T - T_{\infty}) \dots \textcircled{1}$$

$$\frac{d\theta}{dt} + A\theta = B \quad \left[\begin{array}{l} \theta \equiv T - T_{\infty} \\ A = \frac{4h}{\rho C D}, B = \frac{I^2 R_e'}{\rho C \pi D^2 / 4} \end{array} \right]$$

$$\frac{d}{dt} (\theta e^{At}) = B e^{At}$$

$$\theta e^{At} = \frac{B}{A} e^{At} + C$$

$$\theta = T - T_{\infty} = \frac{B}{A} + C e^{-At}$$

$$\underline{T - T_{\infty} - B/A = C e^{-At}} \quad \text{--- } \textcircled{2}$$

At $t=0$, $T = T_i$

$$T_i - T_\infty - \frac{B}{A} = C \quad \text{--- (3)}$$

② ÷ ③

$$\frac{T - T_\infty - B/A}{T_i - T_\infty - B/A} = \exp\left(-\frac{4h}{\rho c D} t\right).$$

$T = 87.7^\circ\text{C}$ (within 1°C of ss temp.), $B/A = 63.7$
 $h = 500 \text{ W/m}^2\text{K}$, $c = 500 \text{ J/kgK}$, $D = 0.001 \text{ m}$

$\Rightarrow \underline{t = 8.3 \text{ s.}}$

Problem 3 – Transient Conduction in a Rising Droplet

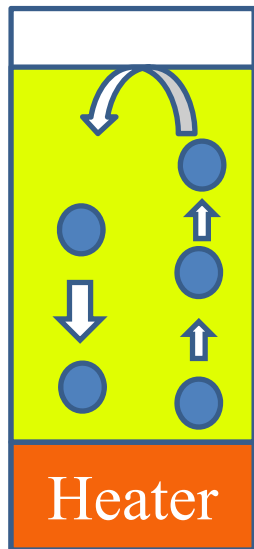
Consider the operation of the "lava" lamp. Polymer is heated at the base and due to buoyant forces begins to rise through a colored liquid in the form of "small" spheres ($r_o = 1.5$ cm). Assume that the colored liquid has a density of $\rho_\infty = 900$ kg/m³ and a uniform temperature of $T_\infty = 325$ K. The polymer sphere has a temperature dependent density given by:

$$\rho = \rho_o - \beta (T - T_\infty), \quad \rho_o = 910 \text{ kg/m}^3, \quad \beta = 2 \text{ kg/m}^3\text{K}$$

If the polymer liquid begins its ascent at a temperature of 350 K, how long before it begins to fall back to be reheated?

Assume: $h = 20$ W/m²K, C_p (sphere) = 400 J/kg K, k (sphere) = 10 W/m K and that the volume and area of the sphere are constant.

The first check should be the validity of LC model - the droplet temperature varies with time (position) but space-wise constant.



The heated droplet will rise and get cooled, increasing its density and therefore at some point it will start its downward journey again.

It is **assumed** that the volume and area of the droplet during this movement remain constant.

The colored liquid temperature is constant.

Check for lumped capacitance validity

$Bi = h r_o / k = (20 \times 0.015)/10 < 0.1$, therefore LC model is valid

$$IN - OUT + GEN = ACCUM$$

$$0 - h A (T - T_{\infty}) + 0 = \frac{d}{dt} (\rho C_p V T)$$

ρ is a function of T and therefore so are V and A of the sphere. However, as mentioned in the problem statement, V and A are assumed to be constant herein.

$$Let \quad \theta \equiv T - T_{\infty}$$

$$-\frac{h A}{C_p V} \theta = \frac{d}{dt} [(\rho_0 - \beta T_\infty) \theta - \beta \theta^2 + \rho_0 T_\infty]$$

$$-\frac{h A}{C_p V} \theta = \left[(\rho_0 - \beta T_\infty) \frac{d\theta}{dt} - 2 \beta \theta \frac{d\theta}{dt} \right]$$

$$-\frac{h A}{C_p V} \int_0^t dt = \left[(\rho_0 - \beta T_\infty) \int_{\theta_0}^{\theta} \frac{d\theta}{\theta} - 2 \beta \int_{\theta_0}^{\theta} d\theta \right]$$

Therefore

$$t = \frac{C_p V}{h A} \left[(\rho_0 - \beta T_\infty) \ln \frac{\theta_0}{\theta} + 2 \beta (\theta - \theta_0) \right]$$

The polymer will begin to fall when its density reached that of the coloured liquid.

The temperature at which it occurs is

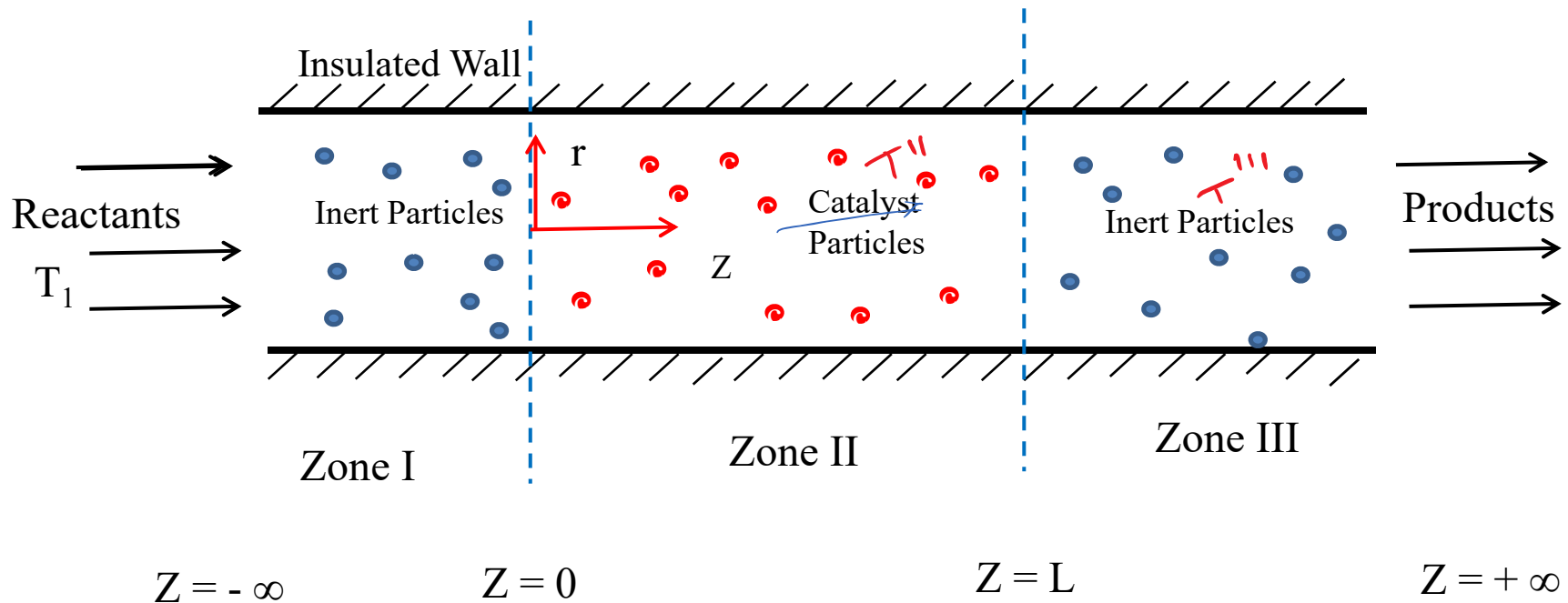
$$900 = 910 - 2 (T - 325) \Rightarrow T = 330 \text{ K}$$

$$\theta_0 = 25 \text{ } ^\circ\text{C}; \theta_{Reversal} = 5 \text{ } ^\circ\text{C}$$

$$t = \frac{400 \times 0.015}{3 \times 20} \left[(910 - 2 \times 325) \ln \frac{25}{5} + 2 \times 2 (25 - 5) \right]$$

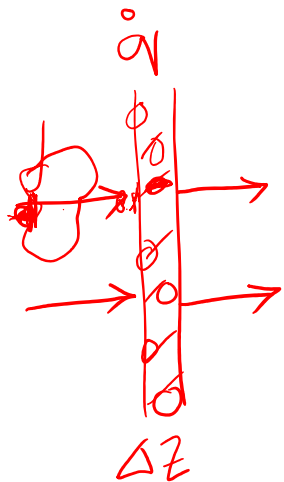
$$t = 498 \text{ s}$$

Heat Conduction with Chemical Heat Source



Superficial Velocity $v_1 = \frac{W}{\pi R^2 \rho_1}$

Volumetric Heat Generation S_C



$$\begin{aligned} & \text{(IN) BY COND } \pi R^2 q_z \Big|_z \Big|_{\text{OUT}} \pi R^2 q_z \Big|_{z+\Delta z} \\ & \text{CONV. } \pi R^2 \rho_1 v_1 c_p (T - T_0) \Big|_z \Big|_{\text{IN}} \pi R^2 \rho_1 v_1 (T - T_0) \Big|_{z+\Delta z} \text{(OUT)} \end{aligned}$$

$$\overset{o}{\underset{Sc}{\text{GEN}}} \rightarrow \pi R^2 \Delta z \overset{o}{\underset{Sc}{S_c}}$$

$$\overset{o}{\text{IN}} - \overset{o}{\text{OUT}} + \overset{o}{\text{GEN}} = 0$$

$$\lim_{\Delta z \rightarrow 0} \left[\frac{q_z \Big|_{z+\Delta z} - q_z \Big|_z}{\Delta z} + \rho_1 v_1 c_p \frac{T \Big|_{z+\Delta z} - T \Big|_z}{\Delta z} \right] = S_c$$

$$\boxed{\frac{dq_z}{dz} + \rho_1 v_1 c_p \frac{dT}{dz} = S_c}$$

$$\left[-k_{z,eff} \frac{d^2 T}{dz^2} + \rho_1 v_1 C_{p,eff} \frac{dT}{dz} = S_c \right]$$

$$\begin{aligned} \text{ZONE 1} &\rightarrow -k_{z,eff} \frac{d^2 T'}{dz^2} + \rho_1 v_1 C_{p,eff} \frac{dT'}{dz} = 0 \\ \text{ZONE 2} &\rightarrow = S_c \\ 3 &\rightarrow = 0 \end{aligned}$$

<u>BC</u>	$z = -\infty$	$T' = T_1$	$\rightarrow$	$k \frac{dT'}{dz} = k \frac{dT''}{dz}$
	$z = 0$	$T' = T''$		
	$z = \infty$	$T''' = \text{FINITE}$		
	$z = L$	$T'' = T'''$	$\rightarrow$	$k \frac{dT''}{dz} = k \frac{dT'''}{dz}$

$$\checkmark \theta^I = C_1 + C_2 e^{Bz}$$

$$\theta'' = C_3 e^{m_3 z} + C_4 e^{m_4 z}$$

$$\rightarrow \theta''' = C_5 + \cancel{C_6 e^{Bz}} \leftarrow$$

$\neq 0$

C_6 as $T'''(\theta''')$
CANNOT BE INFINITE

$$\theta''' = C_5$$

